

Node Structure Visualization

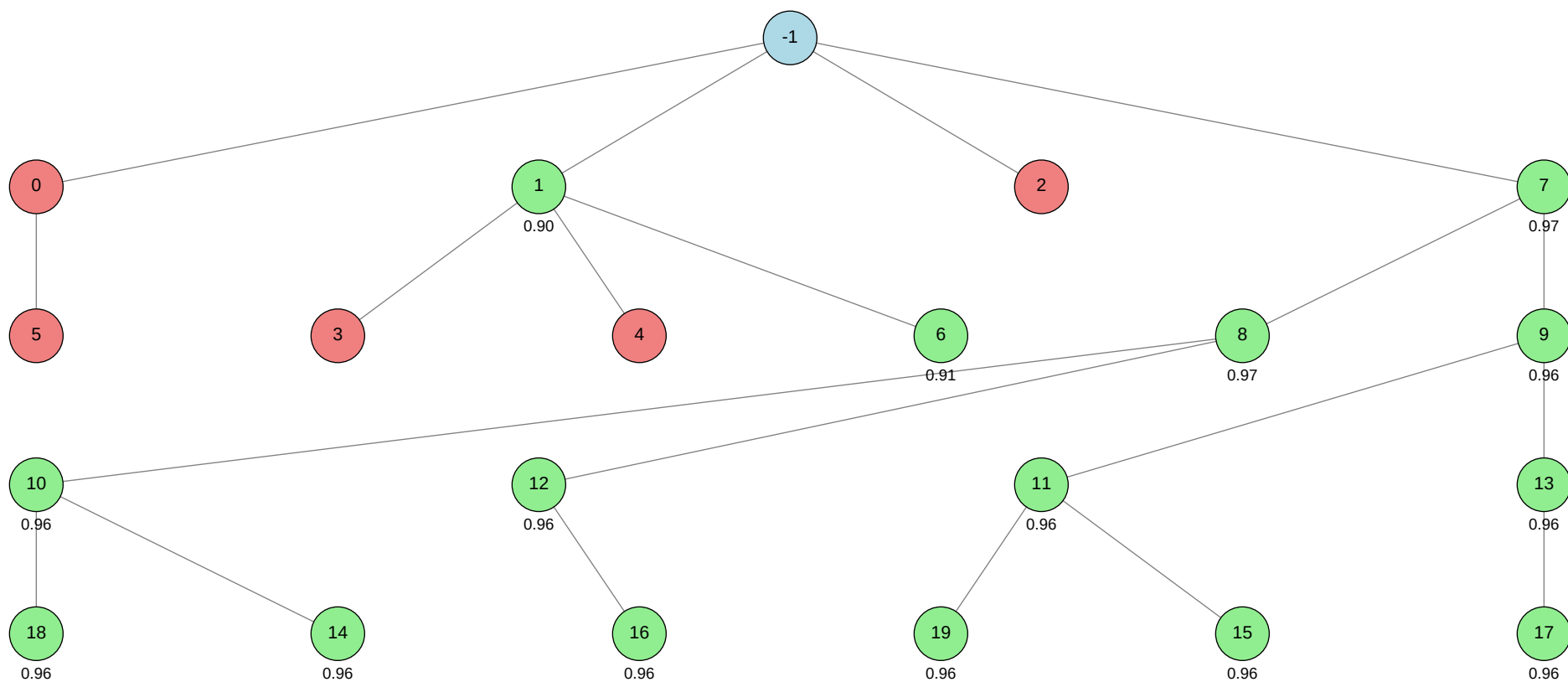
Click on any node in the tree to jump to its detailed information.

Total Nodes: 21 | Best Validation Score: 0.9659 (Node 7)

Node Tree:

Table of Contents:

Node ID	Stage	Status	Tool
-1			
0			autogluon.multimodal
1			machine learning
2			autogluon.tabular
3			machine learning
4			machine learning
5			autogluon.multimodal
6			machine learning
7			autogluon.multimodal
8			autogluon.multimodal
9			autogluon.multimodal
10			autogluon.multimodal
11			autogluon.multimodal
12			autogluon.multimodal
13			autogluon.multimodal
14			autogluon.multimodal
15			autogluon.multimodal
16			autogluon.multimodal
17			autogluon.multimodal
18			autogluon.multimodal
19			autogluon.multimodal



Status	Color
Success	Green
Failure	Red
Neutral	Blue

Node Details:

Node -1 (root)

Property	Value
uct_value	0.5382
ctime	2026-05-20 12:11:58
debug_attempts	0
depth	0
execution_time	0.0
expected_child_count	0
failure_visits	5
id	-1
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✗
num_children	4
prev_tutorial_prompt	None
stage	root
time_step	-1
tool_used	
tools_available	[]
unvalidated_visits	0
validated_reward	14.3040400000000006
validated_visits	15
validation_score	None
visits	20
parent_id	None
child_ids	[0, 1, 2, 7]

Node 0 (evolve)

Property	Value
uct_value	1.7306
ctime	2026-05-20 12:13:14
debug_attempts	1
depth	1
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.394748179489867
failure_penalty	-0.0
failure_visits	2
id	0
is_debug_successful	✗
is_leaf	✗
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	1
prev_tutorial_prompt	None
stage	evolve
time_step	0
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None

visits	2
parent_id	-1
child_ids	[5]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 28-basic_prompt/node_0/conda_env

Resolved 235 packages in 15.67s

Downloading sympy (6.0MiB)

Downloading fonttools (4.8MiB)

Downloading cryptography (4.5MiB)

Downloading hf-xet (4.3MiB)

Downloading timm (2.2MiB)

Downloading pandas (12.2MiB)

Downloading virtualenv (7.2MiB)

Downloading statsmodels (9.9MiB)

Downloading numba (3.6MiB)

Downloading torchvision (7.7MiB)

Downloading numpy (15.6MiB)

Downloading pycryptodome (2.2MiB)

Downloading botocore (14.3MiB)

Downloading lightgbm (3.4MiB)

Downloading nvidia-cuda-cupti-cu12 (9.8MiB)

Downloading ray (69.0MiB)

Downloading plotly (9.4MiB)

Downloading thinc (3.9MiB)

Downloading tokenizers (3.2MiB)

Downloading transformers (11.4MiB)

Downloading py-spy (2.8MiB)

Downloading scipy (34.2MiB)

Downloading triton (162.5MiB)

Downloading spacy (30.8MiB)

Downloading scikit-learn (9.3MiB)

Downloading tensorboard-data-server (6.3MiB)

Downloading pillow (6.3MiB)

Downloading sqlalchemy (3.8MiB)

Downloading matplotlib (8.4MiB)

Downloading nvidia-cublas-cu12 (566.8MiB)

Downloading blis (10.9MiB)
Downloading llvmlite (53.7MiB)
Downloading pyarrow (40.3MiB)
Downloading nvidia-cusparse-cu12 (274.9MiB)
Downloading nvidia-cusparselt-cu12 (273.9MiB)
Downloading torch (858.1MiB)
Downloading xgboost (110.6MiB)
Downloading nvidia-nvjitlink-cu12 (37.4MiB)
Downloading opencv-python (69.5MiB)
Downloading nvidia-cuda-nvrtc-cu12 (84.0MiB)
Downloading tensorboard (5.3MiB)
Downloading catboost (92.7MiB)
Downloading nvidia-curand-cu12 (60.7MiB)
Downloading nvidia-cusolver-cu12 (255.1MiB)
Downloading nvidia-nccl-cu12 (307.4MiB)
Downloading nvidia-cudnn-cu12 (674.0MiB)
Downloading scikit-image (14.1MiB)
Downloading nvidia-cufft-cu12 (184.2MiB)
Downloading nvidia-nvshmem-cu12 (118.9MiB)
Downloading grpcio (6.5MiB)
Downloaded pycryptodo... [truncated, 10725 chars total]

Error Analysis:

ERROR_SUMMARY: The AssertionError occurs because the training data's label column contains three classes: 'benign', 'malignant', and 'non-neoplastic', but the code expects only 'malignant' and 'non-neoplastic'. This mismatch causes the assertion `assert set(classes) == {'malignant', 'non-neoplastic'}` to fail, as the unexpected 'benign' class is present in the data.

SUGGESTED_FIX:

Inspect the train.csv file to confirm the presence of the 'benign' class. Update the data preprocessing step to either filter out rows with the 'benign' label or map 'benign' to 'non-neoplastic' or another appropriate class, depending on the intended binary classification task. Ensure that after preprocessing, the label column contains only 'malignant' and 'non-neoplastic' before proceeding with model training and assertions.

Node 1 (evolve)

Property	Value
uct_value	1.2673
avg_raw_score	0.904285
ctime	2026-05-20 12:21:22
debug_attempts	0
depth	1
execution_time	0.0
expected_child_count	0
exploitation	0.0436472678809417
exploration	0.9098754692839941
failure_penalty	-0.0
failure_visits	2
id	1
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.0872945357618834
num_children	3
prev_tutorial_prompt	None
stage	evolve
time_step	1
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0436472678809417
validated_reward	1.80857

validated_visits	2
validated_weight	0.5
validation_score	0.8984
visits	4
parent_id	-1
child_ids	[3, 4, 6]

Node 2 (evolve)

Property	Value
uct_value	2.4474
ctime	2026-05-20 12:38:27
debug_attempts	0
depth	1
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.2131672923786587
failure_penalty	-0.0
failure_visits	1
id	2
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✗
normalized_failure_visit	0
num_children	0
prev_tutorial_prompt	None
stage	evolve
time_step	2
tool_used	autogluon.tabular
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None

visits	1
parent_id	-1
child_ids	[]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 28-basic_prompt/node_2/conda_env

Resolved 239 packages in 3.62s

Downloading tbb (4.0MiB)

Downloading tcmlib (2.7MiB)

Downloading daal (106.2MiB)

Downloaded tcmlib

Downloaded tbb

Downloading scikit-learn-intelex (4.6MiB)

Downloaded scikit-learn-intelex

Downloaded daal

Prepared 4 packages in 16.01s

Uninstalled 1 package in 1.55s

Installed 237 packages in 35.85s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.1

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260520

+ autogluon-common==1.5.1b20260520

+ autogluon-core==1.5.1b20260520

+ autogluon-features==1.5.1b20260520

+ autogluon-multimodal==1.5.1b20260520

+ autogluon-tabular==1.5.1b20260520

+ autogluon-timeseries==1.5.1b20260520

+ beartype==0.22.9
+ beautifulsoup4==4.14.3
+ blis==1.3.3
+ boto3==1.43.11
+ botocore==1.43.11
+ catalogue==2.0.10
+ catboost==1.2.10
+ certifi==2026.4.22
+ cffi==2.0.0
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ chronos-forecasting==2.2.2
+ click==8.4.0
+ cloudpathlib==0.24.0
+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ daal==2025.9.0
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.... [truncated, 43348 chars total]

Error Analysis:

ERROR_SUMMARY: AutoGluon failed to train any models because the Ray backend (used for parallel model training) attempted to create UNIX domain sockets with file paths exceeding the system limit of 107 bytes. This resulted in repeated OSError exceptions: "validate_socket_filename failed: AF_UNIX path length cannot exceed 107 bytes", causing all model training to be skipped and a final RuntimeError: "No models were trained successfully during fit()".

SUGGESTED_FIX:

Shorten the directory paths used for temporary files and model output so that the full UNIX socket path stays under 107 bytes. Move your working/output directories (e.g., ``/sc-scratch/sc-scratch-ikim-guidlight/be-fair/mlzero/28-basic_prompt/node_2/output``) to a location with a much shorter absolute path (such as ``/tmp/ag/`` or ``/scratch/ag/``), and update all relevant script variables and environment settings to use the new, shorter paths before rerunning the training.

Node 3 (evolve)

Property	Value
uct_value	1.6649
ctime	2026-05-20 12:42:01
debug_attempts	0
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.482079962591042
failure_penalty	-0.0
failure_visits	1
id	3
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
normalized_failure_visit	0
num_children	0
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have then
stage	evolve
time_step	3
tool_used	machine learning

tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	1
parent_id	1
child_ids	[]

Error Message:

```

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `~/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt`
does not contain any dependencies
Using Python 3.11.15 environment at: 28-basic_prompt/node_3/conda_env
Resolved 79 packages in 518ms
Uninstalled 1 package in 1.53s
Installed 78 packages in 31.38s
+ aiohappyeyeballs==2.6.1
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.4.22
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.0
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0

```

- + frozenlist==1.8.0
- + fsspec==2026.2.0
- + h11==0.16.0
- + hf-xet==1.5.0
- + httpcore==1.0.9
- + httpx==0.28.1
- + huggingface-hub==1.15.0
- + idna==3.15
- + jinja2==3.1.6
- + markdown-it-py==4.2.0
- + markupsafe==3.0.3
- + mdurl==0.1.2
- + mpmath==1.3.0
- + multidict==6.7.1
- + multiprocessing==0.70.19
- + networkx==3.6.1
- + numpy==2.4.6
- + nvidia-cublas==13.1.1.3
- + nvidia-cuda-cupti==13.0.85
- + nvidia-cuda-nvrtc==13.0.88
- + nvidia-cuda-runtime==13.0.96
- + nvidia-cudnn-cu13==9.20.0.48
- + nvidia-cufft==12.0.0.61
- + nvidia-cufile==1.15.1.6
- + nvidia-curand==10.4.0.35
- + nvidia-cusolver==12.0.4.66
- + nvidia-cusparselt-cu13==0.8.1
- + nvidia-cusparse==12.6.3.3
- + nvidia-nccl-cu13==2.29.7
- + nvidia-nvjitlink==13.0.88
- + nvidia-nvshmem-cu13==3.4.5
- + nvidia-nvtx==13.0.85
- + opencv-python==4.13.0.92
- + pandas==3.0.3
- + pillow==12.2.0
- + propcache==0.5.2
- + pyarrow==24.0.0
- + pydantic==2.13.4
- + pydantic-core==2.46.4
- + pygments==2.20.0
- + python-calamine==0.6.2
- + python-dateutil==2.9.0.post0
- + pytz==2026.2
- + pyyaml==6.0.3
- + requests==2.34.2

+ rich==15.0.0
- setuptools==82.0.1
+ s... [truncated, 46669 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurs because the version of LightGBM installed in your environment does not support the `early_stopping_rounds` parameter directly in the `LGBMClassifier.fit()` method. This leads to a `TypeError` when the script attempts to pass `early_stopping_rounds` as a keyword argument to `fit()`, causing model training to fail.

SUGGESTED_FIX:

Check the installed LightGBM version (run `pip show lightgbm`) and either upgrade to at least version 3.0.0 (where `early_stopping_rounds` is supported in `fit()`), or remove the `early_stopping_rounds` argument from the `fit()` call and implement early stopping using the `callbacks` parameter instead. After making the appropriate change, rerun the script to ensure model training completes successfully.

Node 4 (evolve)

Property	Value
uct_value	1.6649
ctime	2026-05-20 12:52:36
debug_attempts	0
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.0
exploration	1.482079962591042
failure_penalty	-0.0
failure_visits	1
id	4
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✗
normalized_failure_visit	0
num_children	0
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have them
stage	evolve
time_step	4
tool_used	machine learning

tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	1
parent_id	1
child_ids	[]

Error Message:

```

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `~/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt`
does not contain any dependencies
Using Python 3.11.15 environment at: 28-basic_prompt/node_4/conda_env
Resolved 79 packages in 1.01s
Uninstalled 1 package in 1.58s
Installed 78 packages in 29.53s
+ aiohappyeyeballs==2.6.1
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.4.22
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.0
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0

```

- + frozenlist==1.8.0
- + fsspec==2026.2.0
- + h11==0.16.0
- + hf-xet==1.5.0
- + httpcore==1.0.9
- + httpx==0.28.1
- + huggingface-hub==1.15.0
- + idna==3.15
- + jinja2==3.1.6
- + markdown-it-py==4.2.0
- + markupsafe==3.0.3
- + mdurl==0.1.2
- + mpmath==1.3.0
- + multidict==6.7.1
- + multiprocessing==0.70.19
- + networkx==3.6.1
- + numpy==2.4.6
- + nvidia-cublas==13.1.1.3
- + nvidia-cuda-cupti==13.0.85
- + nvidia-cuda-nvrtc==13.0.88
- + nvidia-cuda-runtime==13.0.96
- + nvidia-cudnn-cu13==9.20.0.48
- + nvidia-cufft==12.0.0.61
- + nvidia-cufile==1.15.1.6
- + nvidia-curand==10.4.0.35
- + nvidia-cusolver==12.0.4.66
- + nvidia-cusparselt-cu13==0.8.1
- + nvidia-cusparse==12.6.3.3
- + nvidia-nccl-cu13==2.29.7
- + nvidia-nvjitlink==13.0.88
- + nvidia-nvshmem-cu13==3.4.5
- + nvidia-nvtx==13.0.85
- + opencv-python==4.13.0.92
- + pandas==3.0.3
- + pillow==12.2.0
- + propcache==0.5.2
- + pyarrow==24.0.0
- + pydantic==2.13.4
- + pydantic-core==2.46.4
- + pygments==2.20.0
- + python-calamine==0.6.2
- + python-dateutil==2.9.0.post0
- + pytz==2026.2
- + pyyaml==6.0.3
- + requests==2.34.2

+ rich==15.0.0
- setuptools==82.0.1
+ s... [truncated, 46137 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurs because the `fit()` method of `LGBMClassifier` does not accept a `verbose` keyword argument in recent LightGBM versions. Verbosity should be controlled via the `callbacks` parameter or by setting the `verbose` parameter in the `LGBMClassifier` constructor, not in `fit()`.

SUGGESTED_FIX:

Remove the `verbose=100` argument from the `clf.fit()` call. If you want logging, set `verbose` in the `LGBMClassifier` constructor or rely on the verbosity provided by the `callbacks` parameter (e.g., `lgb.early_stopping(..., verbose=True)`).

Node 5 (debug)

Property	Value
uct_value	1.1772
ctime	2026-05-20 13:03:14
debug_attempts	1
depth	2
execution_time	0.0
expected_child_count	0
failure_visits	1
id	5
is_debug_successful	✗
is_leaf	✓
is_successful	✗
is_terminal	✓
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	debug
time_step	5
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.0
validated_visits	0
validation_score	None
visits	1
parent_id	0
child_ids	[]

Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 28-basic_prompt/node_5/conda_env

Resolved 235 packages in 3.22s

Uninstalled 1 package in 1.61s

Installed 233 packages in 48.23s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.1

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260520

+ autogluon-common==1.5.1b20260520

+ autogluon-core==1.5.1b20260520

+ autogluon-features==1.5.1b20260520

+ autogluon-multimodal==1.5.1b20260520

+ autogluon-tabular==1.5.1b20260520

+ autogluon-timeseries==1.5.1b20260520

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.11

+ botocore==1.43.11

+ catalogue==2.0.10

+ catboost==1.2.10

+ certifi==2026.4.22

+ cffi==2.0.0

+ chardet==7.4.3

+ charset-normalizer==3.4.7

+ chronos-forecasting==2.2.2

+ click==8.4.0

+ cloudpathlib==0.24.0

+ cloudpickle==3.1.2
+ colorama==0.4.6
+ colorful==0.6.0a1
+ colorlog==6.10.1
+ confection==1.3.3
+ contourpy==1.3.3
+ coreforecast==0.0.16
+ cryptography==48.0.0
+ cycler==0.12.1
+ cymem==2.0.13
+ datasets==4.0.0
+ defusedxml==0.7.1
+ dill==0.3.8
+ distlib==0.4.0
+ einops==0.8.2
+ einx==0.4.3
+ evaluate==0.4.6
+ fastai==2.8.5
+ fastcore==1.8.18
+ fastdownload==0.0.7
+ fastprogress==1.0.5
+ fasttransform==0.0.2
+ filelock==3.29.0
+ fonttools==4.63.0
+ frozendict==2.4.7
+ frozenlist==1.8.0
+ fsspec==2025.3.0
+ fugue==0.9.7
+ future==1.0.0
+ gdown==6.0.0
+ gluonts==0.16.2
+ google-api-core==2.30.3
+ google-auth==2.53.0
+ googleapis-common-protos==1.75.0
+ graphviz==0.21
+ grpcio==1.81.0rc1
+ hf-xet==1.5.0
+ httpx==1.0.dev3
+ huggingface-hub==0.36.2
+ hyperopt==0.2.7
+ idna==3.1... [truncated, 6557 chars total]

Error Analysis:

ERROR_SUMMARY: The error occurs because the MultiModalPredictor constructor in AutoGluon does not accept a seed argument. Passing seed as a keyword argument to MultiModalPredictor.__init__() results in a TypeError, as seed is not a recognized parameter for this method.

SUGGESTED_FIX:

Remove the `seed=RANDOM_SEED` argument from the `MultiModalPredictor` constructor. If reproducibility is required, set the random seed using `numpy`, `random`, and any relevant framework-specific seed-setting functions before model creation, but do not pass seed directly to `MultiModalPredictor`.

Node 6 (debug)

Property	Value
uct_value	1.8394
ctime	2026-05-20 13:07:18
debug_attempts	1
depth	3
execution_time	0.0
expected_child_count	0
failure_visits	0
id	6
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: Intro to Use Machine Learning or Deep Learning algorithms # Condensed: Intro to Use Machine Learning or Deep Learning algorithms Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you # Intro to Use Machine Learning or Deep Learning algorithms You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have them
stage	debug
time_step	6
tool_used	machine learning
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.91018
validated_visits	1

validation_score	0.9102
visits	1
parent_id	1
child_ids	[]

Node 7 (debug)

Property	Value
uct_value	1.6087
avg_raw_score	0.9612616666666668
ctime	2026-05-20 13:19:27
debug_attempts	2
depth	2
execution_time	0.0
expected_child_count	0
exploitation	0.9310183128486124
exploration	0.700422462826868
failure_penalty	-0.0
failure_visits	0
id	7
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9310183128486124
num_children	2
prev_tutorial_prompt	None
stage	debug
time_step	7
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_contribution	0.9310183128486124
validated_reward	12.495470000000003

validated_visits	13
validated_weight	1.0
validation_score	0.9659
visits	13
parent_id	-1
child_ids	[8, 9]

Node 8 (evolve)

Property	Value
uct_value	1.8555
avg_raw_score	0.9612616666666667
ctime	2026-05-20 13:52:55
debug_attempts	0
depth	3
execution_time	0.0
expected_child_count	0
exploitation	0.9310183128486107
exploration	0.9099736640840701
failure_penalty	-0.0
failure_visits	0
id	8
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9310183128486107
num_children	2
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	8
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9310183128486107
validated_reward	5.76757
validated_visits	6
validated_weight	1.0
validation_score	0.9659
visits	6
parent_id	7
child_ids	[10, 12]

Node 9 (evolve)

Property	Value
uct_value	1.8417
avg_raw_score	0.9603300000000001
ctime	2026-05-20 14:26:17
debug_attempts	0
depth	3
execution_time	0.0
expected_child_count	0
exploitation	0.9172219754183345
exploration	0.9968262051089475
failure_penalty	-0.0
failure_visits	0
id	9
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9172219754183345
num_children	2
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	9
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9172219754183345
validated_reward	5.76198
validated_visits	6
validated_weight	1.0
validation_score	0.9603
visits	6
parent_id	7
child_ids	[11, 13]

Node 10 (evolve)

Property	Value
uct_value	2.0100
avg_raw_score	0.96033
ctime	2026-05-20 14:52:03
debug_attempts	0
depth	4
execution_time	0.0
expected_child_count	0
exploitation	0.9172219754183328
exploration	1.2684446626418298
failure_penalty	-0.0
failure_visits	0
id	10
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9172219754183328
num_children	2
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	10
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9172219754183328
validated_reward	2.88099
validated_visits	3
validated_weight	1.0
validation_score	0.9603
visits	3
parent_id	8
child_ids	[18, 14]

Node 11 (evolve)

Property	Value
uct_value	2.0100
avg_raw_score	0.96033
ctime	2026-05-20 15:17:35
debug_attempts	0
depth	4
execution_time	0.0
expected_child_count	0
exploitation	0.9172219754183328
exploration	1.2684446626418298
failure_penalty	-0.0
failure_visits	0
id	11
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9172219754183328
num_children	2
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	11
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9172219754183328
validated_reward	2.88099
validated_visits	3
validated_weight	1.0
validation_score	0.9603
visits	3
parent_id	9
child_ids	[19, 15]

Node 12 (evolve)

Property	Value
uct_value	2.2556
avg_raw_score	0.96033
ctime	2026-05-20 15:43:08
debug_attempts	0
depth	4
execution_time	0.0
expected_child_count	0
exploitation	0.9172219754183328
exploration	1.2684446626418298
failure_penalty	-0.0
failure_visits	0
id	12
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9172219754183328
num_children	1
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	12
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9172219754183328
validated_reward	1.92066
validated_visits	2
validated_weight	1.0
validation_score	0.9603
visits	2
parent_id	8
child_ids	[16]

Node 13 (evolve)

Property	Value
uct_value	2.2556
avg_raw_score	0.96033
ctime	2026-05-20 16:08:51
debug_attempts	0
depth	4
execution_time	0.0
expected_child_count	0
exploitation	0.9172219754183328
exploration	1.2684446626418298
failure_penalty	-0.0
failure_visits	0
id	13
is_debug_successful	✗
is_leaf	✗
is_successful	✓
is_terminal	✗
normalized_failure_visit	0
normalized_score	0.9172219754183328
num_children	1
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	13
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0

validated_contribution	0.9172219754183328
validated_reward	1.92066
validated_visits	2
validated_weight	1.0
validation_score	0.9603
visits	2
parent_id	9
child_ids	[17]

Node 14 (evolve)

Property	Value
uct_value	2.3993
ctime	2026-05-20 16:34:55
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
failure_visits	0
id	14
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	14
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.96033
validated_visits	1
validation_score	0.9603
visits	1
parent_id	10
child_ids	[]

Node 15 (evolve)

Property	Value
uct_value	2.3993
ctime	2026-05-20 17:00:35
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
failure_visits	0
id	15
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	15
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.96033
validated_visits	1
validation_score	0.9603
visits	1
parent_id	11
child_ids	[]

Node 16 (evolve)

Property	Value
uct_value	2.0945
ctime	2026-05-20 17:26:19
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
failure_visits	0
id	16
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	16
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.96033
validated_visits	1
validation_score	0.9603
visits	1
parent_id	12
child_ids	[]

Node 17 (evolve)

Property	Value
uct_value	2.0945
ctime	2026-05-20 17:52:12
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
failure_visits	0
id	17
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	17
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.96033
validated_visits	1
validation_score	0.9603
visits	1
parent_id	13
child_ids	[]

Node 18 (evolve)

Property	Value
uct_value	2.3993
ctime	2026-05-20 18:18:03
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
failure_visits	0
id	18
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	18
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.96033
validated_visits	1
validation_score	0.9603
visits	1
parent_id	10
child_ids	[]

Node 19 (evolve)

Property	Value
uct_value	2.3993
ctime	2026-05-20 18:44:18
debug_attempts	0
depth	5
execution_time	0.0
expected_child_count	0
failure_visits	0
id	19
is_debug_successful	✗
is_leaf	✓
is_successful	✓
is_terminal	✗
num_children	0
prev_tutorial_prompt	### Condensed: ```python # Condensed: ```python Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation
stage	evolve
time_step	19
tool_used	autogluon.multimodal
tools_available	['autogluon.multimodal', 'machine learning', 'autogluon.tabular']
unvalidated_visits	0
validated_reward	0.96033
validated_visits	1
validation_score	0.9603
visits	1
parent_id	11
child_ids	[]

